

A note on *and*, *or* and their Chinese counterparts

The scale formed by *and* and *or*

Truth table for logical connectives

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$
1	1	1	1	1
1	0	0	1	0
0	1	0	1	1
0	0	0	0	1

(Material) implication is the weakest common meaning for *if...then* conditionals

Disjunction: exclusive and inclusive *or*

The agent arrived in Berlin on the 9.25 or the 11.10.

Either you hold the baby or you hold the bags.

Note the exclusive “or” relation which prompts selection among alternatives. Mutual exclusiveness of alternatives is not a logical requirement by the relation of disjunction.

The puzzle

Why is *or* many a time interpreted exclusively rather than inclusively if we assume *or* is the linguistic realization of the disjunction operation?

A pragmatic explanation resorting to scalar implicature!

The informational strength of *and* and *or*

And is more informative than *or*.

In a world w where $p \wedge q$ is true, $p \vee q$ must be true in w .

In a world w where $p \vee q$ is true, $p \wedge q$ may or may not be true.

$\{p \wedge q, p \vee q\}$ can form a scale in terms of information strength, where $p \vee q$ is the weaker term compared with $p \wedge q$.

If the principle of informativeness is assumed, the assertion $p \vee q$ implies the elimination of the stronger proposition $p \wedge q$.

The diverse uses of *and*

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1. conjunction: Lily plays the piano and Helen sings the song.
 2. Sum formation/ group formation:
 - a. I bought both large and small potatoes.
 - b. Mary and Tom came to the party.
 - c. Mary and Tom share the apartment.
 - d. Mary and Tom is a team.
 3. Additivity: 1 and 1 makes 2.

4. Repetition:

a. The hole is getting bigger and bigger.

b. He tried and tried but failed.

5. Differences:

I like city life but there are cities and cities.

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6. Purpose: We will stop and take some photos
 7. Contrast: I love you, and you love her.
 8. Conditional: Work hard, and you will achieve your goal.
 9. Temporality: a. He did some warm-up exercises and (he) started jogging.
b. I went to bed and (I) took off my shoes.
 10. Reformulation: This stadium is huge and it can hold 10,000 audience.
 11. Elaboration: The council built the bridge and John did the steel construction.

12. Causality: a. He tried very hard and finally achieved success. ('result')

b. The television is still hot, and you must have been watched TV. ('induction')

13. Reverse causality/ explanation: John fell and he slipped on a banana skin.

A: Bob wants me to get rid of these mats. He says he trips over them all the time. Still, I don't suppose he'll break his neck.

B: Well, I don't know. JOHN broke his leg and HE tripped on a PERSian RUG

14. Sentential-initial *and*

- a. Introduction: 'We went to the supermarket.' -- 'And what did you buy?'
- b. Addition: Professor White prefers the students who are active in class. And John is such a student.
- c. Topic change: And now the headlines.

Chinese conjunction marker *he* (和) and its disjunctive use

“和”表示选择关系

逃避海关监管将境外【固体废物、液态废物和气态废物】运输进境，情节严重的，处五年以下有期徒刑，并处或者单处罚金；情节特别严重的，处五年以上有期徒刑，并处罚金。

(崔玉珍 2023 “法律语言中的逻辑应用” 复旦大学讲座)

Chinese disjunction marker *huozhe* (或者) and its conjunctive use

“或者”表示并列关系

聚众进行淫乱活动的，对【首要分子或者多次参加的】，处五年以下有期徒刑、拘役或者管制。

(崔玉珍 2023 “法律语言中的逻辑应用” 复旦大学讲座)